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CSC8830: Computer Vision
Module 7

GitHub Repository: https://github.com/yosammy01/CSC8830_COMPUTER_VISION/tree/main/MODULE7

Web Application: https://yosammy01.github.io/CSC8830_COMPUTER_VISION/#/module7

Re-implement and demonstrate on your webpage, the object-size estimation method (Assignment 1) using CALIBRATED STEREO. This means that you must first estimate the distance (Z value) between the point in 3D space and camera and use that to estimate the size of an object.

I. Stereo Calibration

1. I created and printed out a calibration checkerboard image from an online tool. The printout was taped to the wall.
2. I took pictures of the calibration images with the same smart phone camera, from two positions, translated a baseline distance from each other, making sure to maintain a similar distance between the camera and wall, as well as the baseline distance between the two phone views, and changed the orientation of the camera to cover as many angles as possible. A total of 18 pairs of calibration images were obtained.
3. Checkerboard calibration images were placed in the "CSC8830_COMPUTER_VISION\MODULE7\left_images" and "CSC8830_COMPUTER_VISION\MODULE7\right_images" directories. The images with a LEFT view were labeled with "LEFT" at the end of the image name and likewise for the images with the RIGHT view.
4. I executed the script from the terminal: "python stereo_calibration.py"
5. The script loaded the "stereo_calibration_params.json" from module 2 experiment and used it to then generate "stereo_calibration_params.json" with the system's stereo calibration values, which is required by the dimension measurement script, "stereo_object_dimensions.py."

   NOTE: The stereo RMS ideally would be < 1 pixel. I could not achieve this with my manual setup using only 1 camera. So, the results using this calibration will have more than ideal error.

**Stereo Calibration Quality Metrics**

| Metric | Value | Notes |
|---|---|---|
| Total Image Pairs | 18 | Left/Right image sets collected in 'left_images/' and 'right_images/' |
| Valid Pairs Used | 17 | Pair 5 excluded (right image checkerboard not detected) |
| Stereo RMS Reprojection Error | 18.246857 pixels | Warning: High error. Ideal is < 1.0 pixel. |
| Estimated Baseline (\|\|T\|\|) | 343.45 mm | Physical distance between optical centers (34.35 cm) |

| | | | | |
|---|---|---|---|---|
| **Camera Matrix (K)** | | | | |
| 3110.9584 | 0.0000 | 1473.5247 | | |
| 0.0000 | 3106.8560 | 2007.9239 | | |
| 0.0000 | 0.0000 | 1.0000 | | |
| | | | | |
| **Distortion Coefficients (dist)** | | | | |
| -0.0142 | 0.2687 | 0.0014 | -7.845E-04 | -0.4294 |
| | | | | |
| **Rotation Matrix (R) [Left -> Right]** | | | | |
| 0.9938 | -0.0369 | 0.1049 | | |
| 0.0371 | 0.9993 | 5.774E-06 | | |
| -0.1049 | 0.0039 | 0.9945 | | |
| | | | | |
| **Translation Vector (T in mm) [Left -> Right]** | | | | |
| -343.0559 | | | | |
| 0.2925 | | | | |
| 16.4516 | | | | |
| | | | | |
| **Essential Matrix (E)** | | | | |
| -0.6413 | -16.4391 | 0.2907 | | |
| -19.6255 | 0.7270 | 342.8882 | | |
| -13.0246 | -342.8087 | -0.0327 | | |

| | | | |
|---|---|---|---|
| **Fundamental Matrix (F)** | | | |
| 1.343E-08 | 3.448E-07 | -7.311E-04 | |
| 4.116E-07 | -1.527E-08 | -0.0229 | |
| 2.418E-06 | 0.0219 | 1.0000 | |
| | | | |
| **Left Projection Matrix (P1)** | | | |
| 3761.5695 | 0.0000 | 1442.7450 | 0.0000 |
| 0.0000 | 3761.5695 | 2012.3283 | 0.0000 |
| 0.0000 | 0.0000 | 1.0000 | 0.0000 |
| | | | |
| **Right Projection Matrix (P2)** | | | |
| 3761.5695 | 0.0000 | 1442.7450 | -1.292E+06 |
| 0.0000 | 3761.5695 | 2012.3283 | 0.0000 |
| 0.0000 | 0.0000 | 1.0000 | 0.0000 |
| | | | |
| **Disparity-to-Depth Matrix (Q)** | | | |
| 1.0000 | 0.0000 | 0.0000 | -1442.7450 |
| 0.0000 | 1.0000 | 0.0000 | -2012.3283 |
| 0.0000 | 0.0000 | 0.0000 | 3761.5695 |
| 0.0000 | 0.0000 | 0.0029 | 0.0000 |

II. Stereo Image Procurement and Methodology

1. Using the known computer vision formulas and the $f_x$ and $f_y$ from the Intrinsic matrix, the real-world dimensions of the objects could be found:

$$X_{world} = (X_{im} * Z_{world}) / f_x$$

$$Y_{world} = (Y_{im} * Z_{world}) / f_y$$

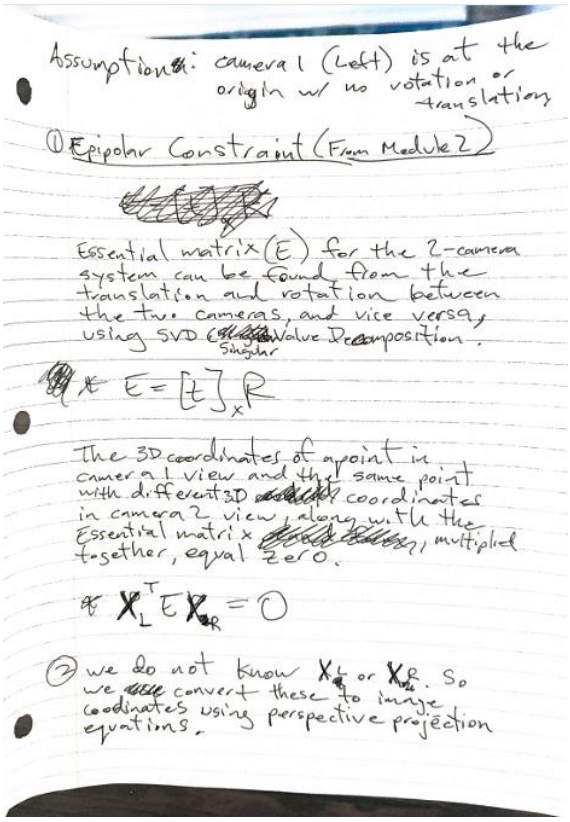
2. The $Z_{world}$ variable is the unknown variable for this experiment, unlike module 2, where we measured and hardcoded that value in the script. The calibrated stereo method calculates this value.
3. The LEFT and RIGHT views of the objects were obtained with the same camera, in a similar setup that was used to take the calibration images. The baseline distance between the camera views was consistent.

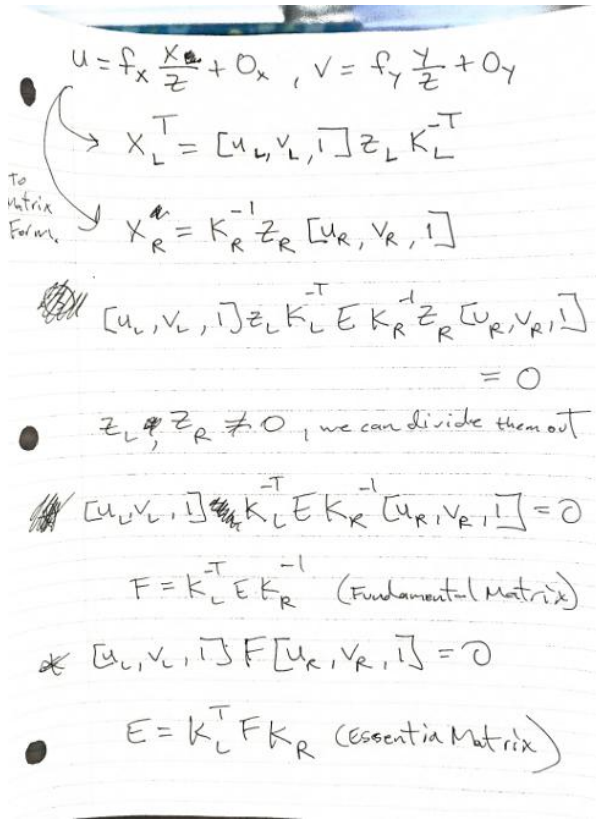
III. Implementation of a python script to find the real world 2D dimensions of an object using the calibrated stereo method. Validation was done for 20 different object measurements – the same objects from module 2. The error estimate statistics were obtained.

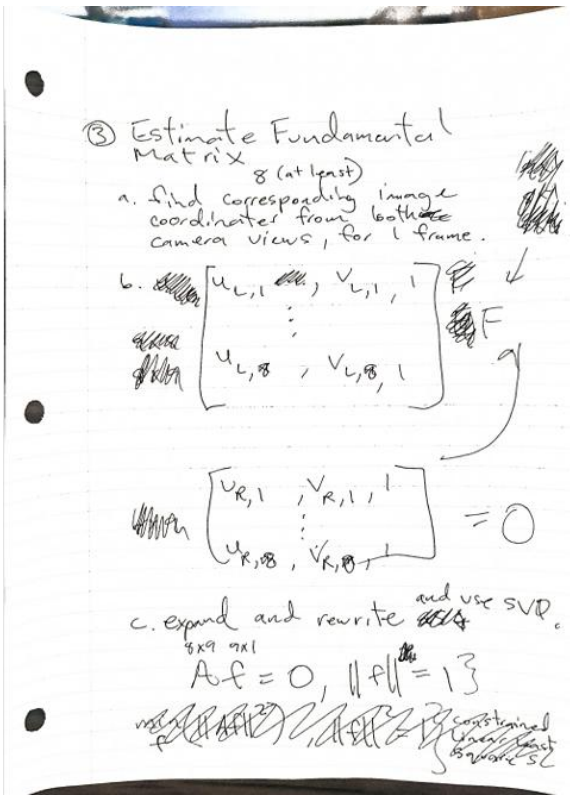
1. The object images were placed in the "CSC8830_COMPUTER_VISION\MODULE7\stereo_measurements_left" and "CSC8830_COMPUTER_VISION\MODULE7\stereo_measurements_right" directories. The images with a LEFT view were labeled with "LEFT" at the end of the image name and likewise for the images with the RIGHT view.
2. Executed the script from the terminal: "python stereo_object_dimensions.py"
3. The script displayed bounding boxes over the detected objects. I pressed any key to proceed to the next image.
4. A "stereo_dimensions_output.txt" was produced by the script, with the estimated real-world dimensions of each object detected in the 20 measurement images.

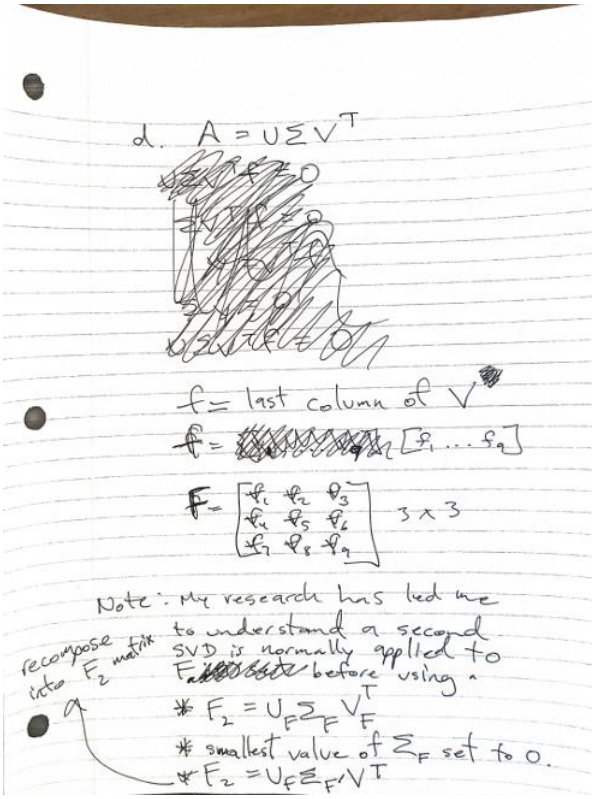
| Object Name | Disparity (px) | Depth Z (mm) | Estimated Width (mm) | Actual Width (mm) | Width Error (%) | Estimated Height (mm) | Actual Height (mm) | Height Error (%) |
|---|---|---|---|---|---|---|---|---|
| air_purifier | 442.31 | 1368.68 | 259.57 | 282.60 | 8.15% | 363.88 | 381.00 | 4.49% |
| black_box | 429.91 | 1382.45 | 204.86 | 230.20 | 11.01% | 254.52 | 277.80 | 8.38% |
| book | 352.23 | 1542.25 | 106.09 | 112.70 | 5.87% | 169.27 | 177.80 | 4.80% |
| box | 398.18 | 1453.23 | 235.90 | 254.00 | 7.13% | 331.63 | 336.50 | 1.45% |
| brick | 394.07 | 1447.57 | 176.82 | 203.20 | 12.98% | 88.99 | 92.07 | 3.35% |
| catalog_case | 410.42 | 1385.90 | 463.31 | 457.20 | 1.34% | 365.78 | 379.41 | 3.59% |
| clock_radio | 382.81 | 1465.89 | 96.60 | 107.90 | 10.47% | 122.67 | 127.00 | 3.41% |
| large_circle_lid | 356.66 | 1526.53 | 174.44 | 190.50 | 8.43% | 174.44 | 190.50 | 8.43% |
| large_rectangle_lid | 380.75 | 1471.03 | 215.15 | 247.60 | 13.11% | 184.18 | 225.40 | 18.29% |
| remote_control | 373.40 | 1490.18 | 145.62 | 169.86 | 14.27% | 46.53 | 49.21 | 5.45% |
| rice_cooker | 395.21 | 1444.98 | 163.96 | 214.31 | 23.49% | 129.30 | 157.16 | 17.73% |
| sandal | 374.21 | 1486.02 | 256.03 | 295.27 | 13.29% | 92.31 | 95.25 | 3.09% |
| shoe_box | 379.71 | 1481.81 | 300.56 | 330.20 | 8.98% | 129.25 | 114.30 | 13.08% |
| small_rectangle_lid | 343.35 | 1538.27 | 169.11 | 206.40 | 18.07% | 133.19 | 133.30 | 0.08% |
| small_round_lid | 387.86 | 1457.09 | 129.74 | 152.40 | 14.87% | 129.74 | 152.40 | 14.87% |
| small_safe | 379.44 | 1463.90 | 290.81 | 323.85 | 10.20% | 244.07 | 254.00 | 3.91% |
| smart_phone | 332.12 | 1562.31 | 68.80 | 79.38 | 13.33% | 150.35 | 158.75 | 5.29% |
| speaker_box | 404.30 | 1430.13 | 184.34 | 206.38 | 10.68% | 182.28 | 214.31 | 14.95% |
| sponge | 326.39 | 1595.15 | 94.35 | 104.77 | 9.95% | 58.02 | 61.91 | 6.28% |
| table_top | 403.43 | 1411.73 | 425.66 | 481.00 | 11.51% | 326.71 | 368.30 | 11.29% |

Derive the procedure for estimating the size of the object in Problem 1 using UNCALIBRATED STEREO. Show all calculations and derivations.


Assumption: camera 1 (Left) is at the origin w/ no rotation or translation
① Epipolar Constraint (From Module 2)
Essential matrix (E) for the 2-camera system can be found from the translation and rotation between the two cameras, and vice versa, using SVD (Singular Value Decomposition).
$E = [t]_x R$
The 3D coordinates of a point in camera 1 view and the same point with different 3D coordinates in camera 2 view, along with the Essential matrix, multiplied together, equal zero.
$X_L^T E X_R = 0$
② we do not know $X_L$ or $X_R$. So we convert these to image coordinates using perspective projection equations.


$u = f_x \frac{X}{Z} + O_x$, $v = f_y \frac{Y}{Z} + O_y$
To Matrix Form
$X_L^T = [u_L, v_L, 1] z_L K_L^{-T}$
$X_R = K_R^{-1} z_R [u_R, v_R, 1]$
$[u_L, v_L, 1] z_L K_L^{-T} E K_R^{-1} z_R [u_R, v_R, 1] = 0$
$z_L, z_R \neq 0$, we can divide them out
$[u_L, v_L, 1] K_L^{-T} E K_R^{-1} [u_R, v_R, 1] = 0$
$F = K_L^{-T} E K_R^{-1}$ (Fundamental Matrix)
$[u_L, v_L, 1] F [u_R, v_R, 1] = 0$
$E = K_L^T F K_R$ (Essential Matrix)


③ Estimate Fundamental Matrix
8 (at least)
a. find corresponding image coordinates from both camera views, for 1 frame.
b.
F
c. expand and rewrite and use SVD.
8x9 9x1
$Af = 0, \|f\| = 1$
constrained linear least squares


d. $A = U \Sigma V^T$
f = last column of V
$f = [f_1 \ldots f_9]$
$F = \begin{bmatrix} f_1 & f_2 & f_3 \\ f_4 & f_5 & f_6 \\ f_7 & f_8 & f_9 \end{bmatrix}$ 3x3
Note: My research has led me to understand a second SVD is normally applied to F before using it to recompose first into $F_2$ matrix
* $F_2 = U_F \Sigma_F V_F^T$
* smallest value of $\Sigma_F$ set to 0.
* $F_2 = U_F \Sigma_{F'} V^T$

(4)

$P_L = [I|0]$ } camera 1 (Left) at origin

$P_R = [[e_2]_x F | e_2]$

$e_2$ = projected center of Left camera, seen in Right image

$F^T e_2 = 0$ } use SVD

$F = U \Sigma V^T$

$e_2$ = last column of U

$[e_2]_x = \begin{bmatrix} 0 & -e_z & e_y \\ e_z & 0 & -e_x \\ -e_y & e_x & 0 \end{bmatrix}$

(5) Cross product of $u_i$ and point $X_p$ is zero

$u_L \times (u_L X_p) = 0$

$u_R \times (u_R X_p) = 0$

expand and reform

$AX_p = 0$

use SVD.

$A = U \Sigma V^T$

$X_p$ = last column of V

(6) $X_c \approx H X_p$

expand + form new matrix A

$Ah = 0$

use SVD, $A = U \Sigma V^T$

h = last column of V

$H = \begin{bmatrix} h_1 & h_2 & h_3 & h_4 \\ h_5 & h_6 & h_7 & h_8 \\ h_9 & h_{10} & h_{11} & h_{12} \\ h_{13} & h_{14} & h_{15} & h_{16} \end{bmatrix}$

$X_{E,target} = HX_p$

$= [X_w, Y_w, Z_w, W_w]^T$

(7) Real world Coordinates

$X_{world} = \frac{X_w}{W_w}$

$Y_{world} = \frac{Y_w}{W_w}$

$Z_{world} = \frac{Z_w}{W_w}$

(8) using euclidean distance between two points, we can find dimensions

$d = \sqrt{(X_B - X_A)^2 + (Y_B - Y_A)^2 + (Z_B - Z_A)^2}$

Implement an application to find the dimensions of a room in a house using videos. You can use stereo setups.

I. Video Procurement and Methodology

1. Using the same setup as used for the calibrated stereo object dimension estimating experiment, I took video footage of a wall from my outside patio. I was not able to use interior room. So, I am assuming the dimensions of a square room can be obtained by finding the dimensions of one of the walls. From there, the other dimensions can be found. So, this one wall of my patio can be assumed to be a wall of an enclosed room.
2. The $Z_{world}$ variable is the unknown variable for this experiment, unlike module 2, where we measured and hardcoded that value in the script. The calibrated stereo method calculates this value.
3. The LEFT and RIGHT videos of the wall were obtained with the same camera, in a similar setup that was used to take the calibration images. The baseline distance between the camera views was consistent. The videos are 6 seconds long and the camera was stationary for each video.

   https://youtube.com/shorts/W1lUVtJKRtM

   https://youtube.com/shorts/T6qNHnZRI2c

II. Implementation

1. The stereo videos were placed in the "CSC8830_COMPUTER_VISION\MODULE7\room_videos_left" and "CSC8830_COMPUTER_VISION\MODULE7\room_videos_right" directories. The video with a LEFT view was labeled with "LEFT" at the end of the video name and likewise for the video with the RIGHT view.
2. Executed the script from the terminal: "stereo_room_dimensions.py"
3. The script displayed a bounding box over the detected wall in the video frame samples.
4. A "room_dimensions_output.txt" was produced by the script, with the estimated real-world dimensions of the room.

| STEREO ROOM DIMENSION ESTIMATION REPORT | | | | |
|---|---|---|---|---|
| | | | | |
| 1. VIDEO & CALIBRATION METADATA | | | | |
| Left View Video | MODULE7/room_videos_left/video_LEFT.mp4 | | | |
| Right View Video | MODULE7/room_videos_right/video_RIGHT.mp4 | | | |
| Native Resolution | 1080x1920 | | | |
| Native FPS | 59.85 | | | |
| Total Frames | 360 | | | |
| Baseline | 343.45 mm (34.35 cm) | | | |
| WLS Disparity Filter | Initialized successfully | | | |
| Focal Lengths | fx=1022.1, fy=1022.1 | | | |
| Processing Time | 87.81s (2.0 FPS avg) | | | |
| | | | | |
| 2. ESTIMATED VS ACTUAL ROOM DIMENSIONS | | | | |
| Dimension | Estimated Value | Actual Value | Absolute Error | Percentage Error |
| Room Width | 2.528 m (252.8 cm \| 8 ft 3.5 in) (std: 0.004 m) | 2.965 m (296.5 cm \| 9 ft 8.8 in) | 0.437 m | 14.75% |
| Room Height | 1.294 m (129.4 cm \| 4 ft 2.9 in) (std: 0.004 m) | 2.146 m (214.6 cm \| 7 ft 0.5 in) | 0.852 m | 39.71% |
| Room Length (Depth) | 3.250 m (325.0 cm \| 10 ft 8.0 in) (std: 0.000 m) | 2.965 m (296.5 cm \| 9 ft 8.8 in) | 0.285 m | 9.60% |
| Floor Area | 8.22 m^2 (88.4 sq ft) | 8.79 m^2 (94.7 sq ft) | 0.57 m^2 | 6.53% |
| Room Volume | 10.63 m^3 (375.4 cu ft) | 18.87 m^3 (666.5 cu ft) | 8.24 m^3 | 43.68% |